

Q Find the λ -closure of a set of states for each state of the given NFA lambda in Figure-1. (3 Marks)

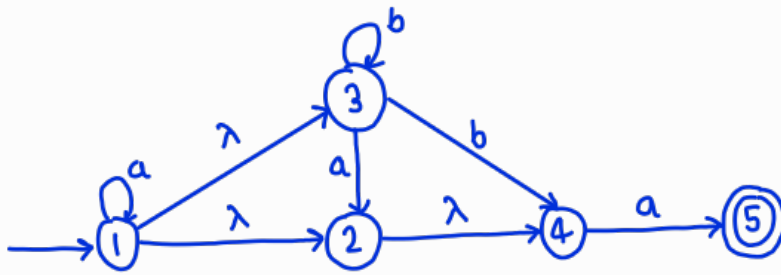


Figure-1

⇒ λ -closure of a state is the set of states reachable from that state following only epsilon (ϵ) transitions, including the state itself.

set of states = $\{1, 2, 3, 4, 5\}$

for each state, find all states reachable by ϵ -transitions / λ -transitions.

$$\lambda\text{-closure}(1) = \{1, 2, 4, 3\}$$

$$\lambda\text{-closure}(2) = \{2, 4\}$$

$$\lambda\text{-closure}(3) = \{3\}$$

$$\lambda\text{-closure}(4) = \{4\}$$

$$\lambda\text{-closure}(5) = \{5\}$$